

Methods 3: Multilevel Statistical Modeling and Machine Learning

Week 50: *Evaluation results*

December 9, 2025

The course plan

Week 37:

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model (Lau)

Week 38:

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects? (Lydia)

Week 39:

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous? (Lau)

Week 40:

Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity

Class 3: Code review (Lydia)

Week 41:

Lesson 4: Explanation versus prediction

Class 4: Summarising mixed models (Lydia)

Week 42:

no teaching (go fetch potatoes instead)

Week 43:

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running (Lau)

Week 44:

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses (Lau)

Week 45:

Lesson 7: Model evaluation and hyperparameter tuning

Class 7: Code review (Lydia)

Week 46:

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data? (Lau)

Week 47:

Lesson 9: Neural networks

Class 9: Code review (Lau)

Week 48:

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource) (Lydia)

Week 49:

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning (Lydia)

Week 50:

Student presentations:

Class 12: Work on portfolios: Ask anything! (Lydia & Lau)

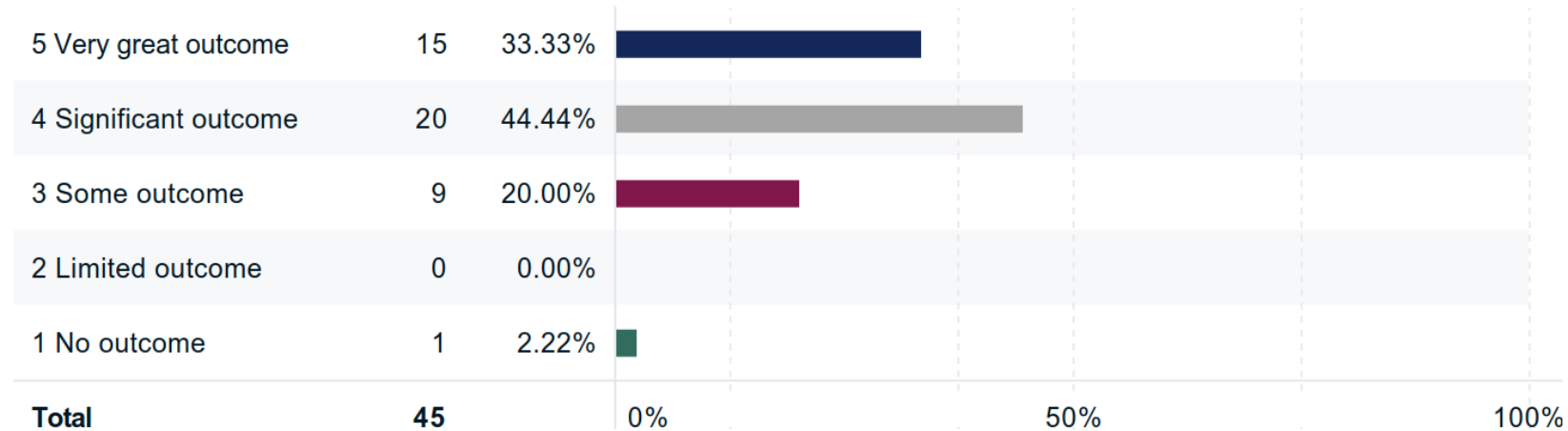
Response rate

	Student
Responded	45
Invited	66
Response Ratio	68.18%

Likert scale 1-5

	Mean
I rate the overall outcome of the course as	4.07
I see how the course is relevant to my overall education	4.82
The course has contributed to building my academic knowledge and competences	4.56
This course has motivated me to engage in class and work actively with the content	4.13
I think the dialogue between teacher(s) and students has been constructive	4.40
I think the learning environment has been safe	4.50

I rate the overall outcome of the course as

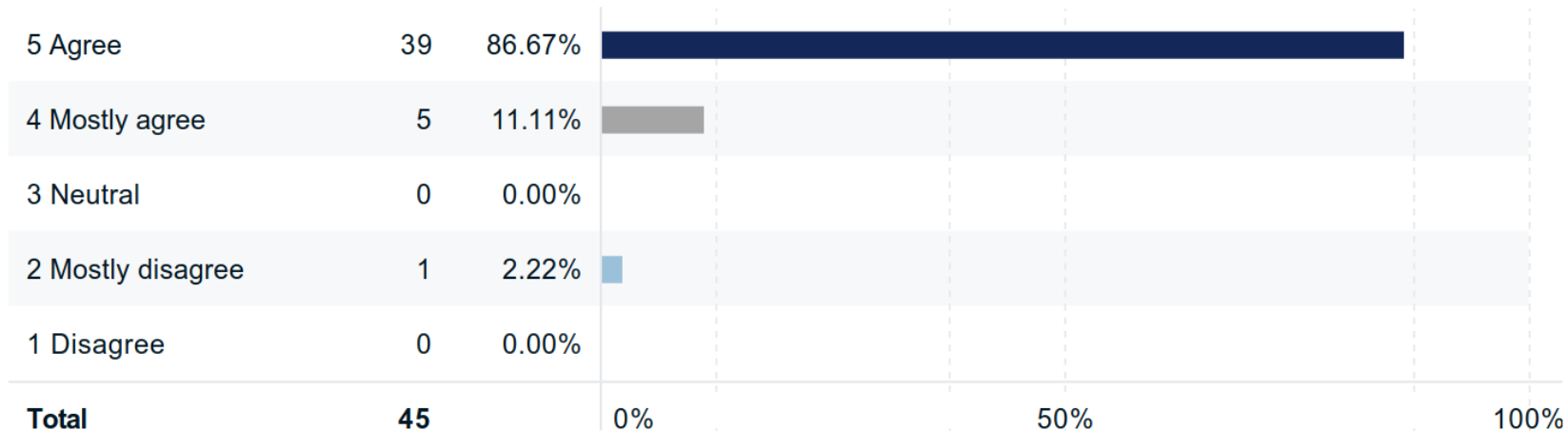


Statistics	Value
Mean	4.07

The course includes instruction on statistical techniques relying on the Generalised Linear Model, including multilevel modelling and basic machine learning procedures (e.g. predictive approaches and validation techniques).

Comment
- It's A LOT of information packed into very few lectures. Especially for the machine learning part. It's nice to get an introduction to this stuff, but it's by no means a comprehensive introduction to use of machine learning for statistics. When you pack in a lot of information it's hard to retain and it's very hard to become skillful at doing either GLMs or machine learning because we have no time to properly integrate the information we've learned.
- The concepts were quite hard to understand as they were hard to relate the models we learned about in class, to topics we would actually use them for. The machine learning part had a lot of good examples which made it easier to follow (ex. breast cancer, flower pedal shapes). However, the topics we learned in the middle of the course were a bit harder to grasp as I wouldn't necessarily know when to apply which model.
- It would be nice to spend more time on some of the underlying math instead of jumping directly into machine learning.
- Jeg tror, at mit udbytte ville være endnu større, hvis jeg selv havde lagt flere kræfter i faget
- Jeg føler at materialet er svært, så det er ikke alt jeg nåede at forstå, men jeg føler også at jeg har lært meget.
- Loved every bit of the course! Would've liked to get more hands-on practice with more of the stuff we've been working on. I think shorter assignments (especially assignment_2 was long and tedious) and more in-class exercises (akin to those in Methods 1) would've allowed us to cover more topics and get more experience working with the course material.
- The course presented very useful, paramount methods for data analysis.

I see how the course is relevant to my overall education

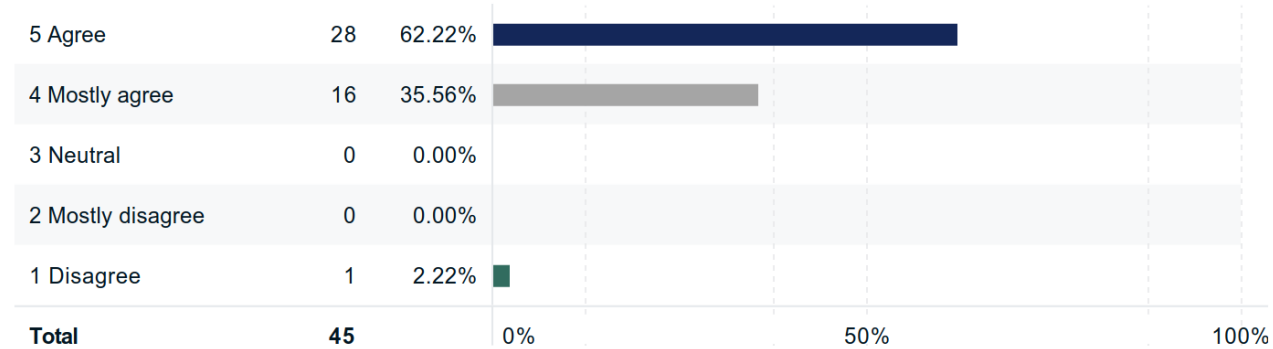


Statistics	Value
Mean	4.82

Comment

- Many good tools, which can also be used in other courses
- I really enjoyed being introduced more in depth to different kinds of data distributions and how to do analysis with these which is a great tool for both other courses but also for real-life applications
- Det er virkelig relevant, så jeg vil faktisk ønske at der var sat mere tid af til Methods (fx endnu en holdtime om ugen). På den måde kunne jeg måske have nået at forstå det hele bedre.
- I think it clearly connects to the other courses.
- Great to be able to learn about and practice GLMM and ML stuff! Would've maybe felt more confident if we got more in-depth knowledge/practice doing the maths behind the material.
- Very relevant! Now I have my swiss knife even bigger.

The course has contributed to building my academic knowledge and competences



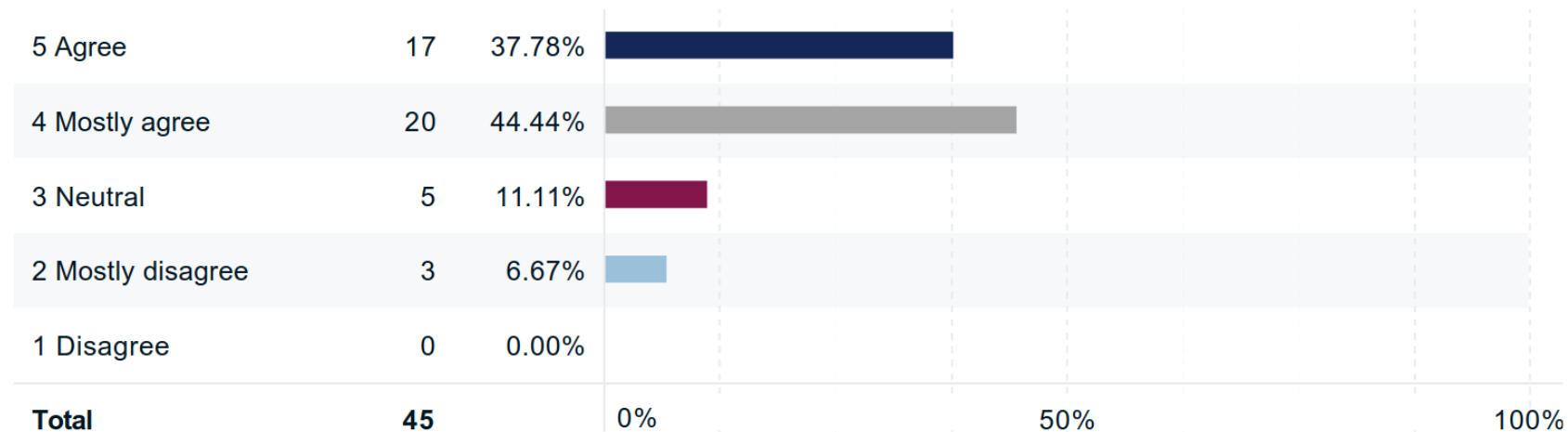
Statistics	Value
Mean	4.56

Comment

- Ja, jeg føler at jeg har lært en masse, selvom jeg ikke nåede at forstå det hele.

- Everything seems to be very relevant to our programme going forward.

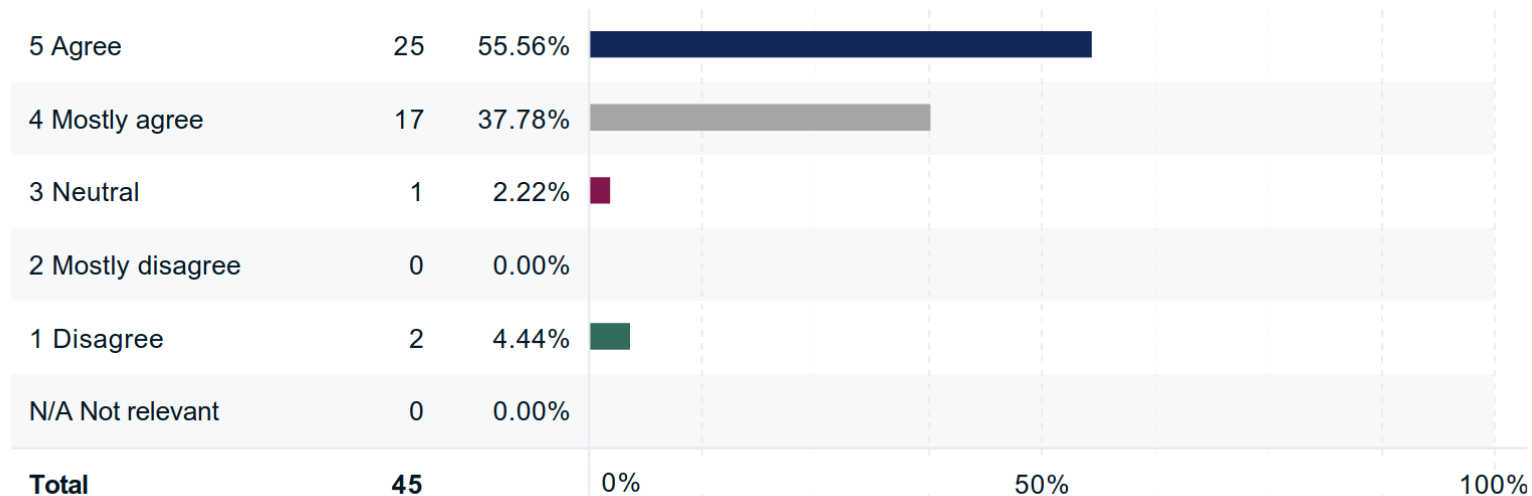
This course has motivated me to engage in class and work actively with the content



Statistics	Value
Mean	4.13

Comment
- The assignments are good at doing this!!
- <u>Getting the slides</u> before lectures would have been a great help in engaging actively in the content. I know there has been some debate whether it would be beneficial or not, but if basically the whole course would like the slides before the lectures it would be helpful to listen to that feedback.
- More class exercises and not just assignments like in methods 1 and 2 would be nice to get a more hands on experience with the concepts and math.
- <u>Live coding</u> fungerede ikke for mig.
- I think even though some concepts were confusing I was still very motivated to learn even if it meant a lot of trial and a lot of error
- Jeg vil tilgængæld også nævne at der har været tidspunkter til <u>live coding</u> til forelæsninger som har været for svært for mig at følge med i fordi det har været for svært ren kode mæssigt
- Sometimes it could feel a little overwhelming with all the tough math, but for the most part is has motivated me.
- Too much focus on the assignments. Would've preferred to have more in-class exercises covering more topics than long (and sometimes vaguely written) assignments, especially assignment_2 did not explain well enough why we did the things we were asked to do.
- Undervisningen har været på meget højt niveau, hvorfor jeg har oplevet at bruge rigtigt meget tid på kurset. Andet, har ikke rigtigt været en mulighed, hvis man gerne vil kunne følge med
- My biggest motivation were the assignments, so I would say that they could be shorter but dedicated to every topic we study.

I think the dialogue between teacher(s) and students has been constructive



Statistics	Value
Mean	4.40

Comment

- Lau seems open to feedback, and he wants people to talk and discuss subjects and class structure

- Virkelig nice mængde af assignments!!! Fedt at du lige har justeret i forhold til sidste år, fire assignments gav rigtig god mening

- Er ikke så vild med harbonering, men forstår også godt meningen med det, men føler bare, det gjorde mig lidt nervøs og ikke lyst til at engagere.

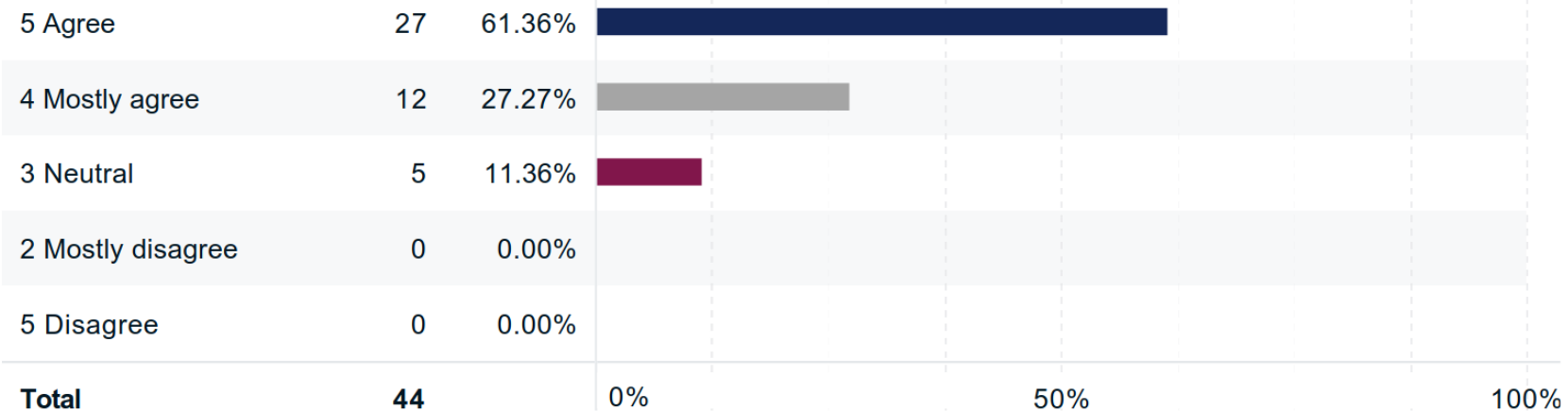
- De første lectures skulle man lige tune ind på hvordan det er at have en ny underviser men især portfolio opgaverne om machine learning var virkelig godt kommunikeret.

- Both Lau and Lydia are great at explaining concepts through different angles to make the topics more intuitive and understandable for everyone

- Det kunne være skønt med mere feedback. Måske kunne man lave nogle opgaver i holdtimerne, som man senere kunne se løsningen på.

- Good: Lau is an amazing lecturer, and he is very approachable when you have questions. I also likes that he insists that there are no stupid questions, and that he corrects himself whenever he makes mistakes. Bad: I don't think that uploading slides after the lecture has the effect Lau intends. In other Methods courses (1 and 2), I loved having a Padlet to ask questions. However, it doesn't seem that Lau used his CryptPad very much – and when he did, his responses seemed a bit rushed and weren't always helpful.

I think the learning environment has been safe



Statistics

Statistics	Value
Mean	4.50

Comment

- some people might find it uncomfortable to speak in front of everyone during the lecture, so asking people randomly can be a bit challenging, but I think the opportunity to always be able to pass the answer on was very helpful, and well accepted by lau!!
- Jeg synes, det er godt, at Lau bare stiller os spørgsmål direkte, uden at vi nødvendigvis har hånden oppe – så tvinger det lidt alle til at svare
- Jeg forstår at Lau har høje forventninger til os og der er klart en svær at finde en balance mellem at udfordre os og ikke stresse os for meget. Det med at kalde os ud på navn i forelæsningsen har til tider stresset mig, fordi jeg har haft svært ved at forstå nogle af teksterne. Det har været rart når det har været to og to, men ja jeg har nogle gange frygtet forelæsningsen.
- Sometimes it could create a little bit of nervousness because you know that Lau might ask you during a lecture without you raising your hand, i see why this is helpful, but also a bit nerve-racking.
- Definitely – it has been a good learning environment, where I've felt more confident about asking questions if in doubt.
- Good: Lau is doing great by pointing out that we can always interrupt with questions (and he even means it!). He also always seems happy to answer questions during breaks and stuff like that. Bad: Picking out students at random is not conducive to a safe learning environment.

Which elements of the course have contributed most to your learning?

Comment

Lectures and assignments

The assignments. I wish there were more exercises to hone our stats skills and that there was more of a building of skills. Assignment 2 felt like being thrown into a very deep pool of statistics with no swimming training. I know there were more assignments – maybe not assignments but I think exercises underway to build up to assignments is great. The way we did it in methods 02 was fantastic.

Comment
The combination/balance of theory on lectures and coding on classes. Workload was ideal, was not too easy but could deal with assignment for a good amount of time which was great. Would have been way better and easier if Methods 2 dealt with calculus, optimization and linear algebra more deeply. Still managed but an un unusual amount of Youtube videos needed to be watched if one wanted to understand the underlying mathematics, which is okay but Methods 2 feels a bit "wasted" looking back given it could provide a strong math basis that helps later on.
In lecture coding parts
Classes have been a great help, especially talking about your code with people outside of your own study group. Lydia has also been really good at breaking down complicated concepts into more digestable ones.
lectures and assignments
Assignments
The practical exercises have been most useful to understanding how to apply the theories introduced in lectures. There has been a very good balance between classes and lectures.
the assignments
Lau
The workbooks
I really liked how the readings for machine learning was closely correlated to what we learned in class. It helped alot, the python book was very good. The assignments helped alot too. Maybe a little bit clearer formulation could make them even better
the interactiveness of the lectures and the lecturer's supportive approach to our questions

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Portolio assignments

Machine learning delen og det at skrive i python

Video explanations/visualizations of concepts, exercises/assignments getting hands on practice, classes with time to go over the content of the lecture again because it always helps to hear it multiple times

The assignments! and also that we have had time in class to work on them with help from Lau and Lydia. When we have watched videos and graphics during the lectures. I think it really helps to have it visualised. I think the amount of assignments was good as they were quite long.

The lectures have been really well-structured, precise, and easy to follow. Furthermore, the readings have been very relevant for the content in the lectures. And Lydia is an amazing instructor. I think the amount of assignments are adequate.

Assignments and also that Lau asks us directly in class

En blanding af lectures og classes, så det er har virkelig været et godt fag – uden tvivl meget mere spændende og informativt end methods 1&2.

Afleveringerne har faktisk været rigtig brugbare til at forstå, hvad vi egentlig lavede. Jeg synes personligt, at forelæsningerne har været ret svære at følge med i og teksterne har også været ret svære. YouTube videoer har været gode. Jeg synes mange exercises i afleveringerne har været unødvendigt kompliceret formuleret. Kunne godt forsimples.

Getting the hands on practise with the topics, through assignments or exercises. Including exercise time in the lectures was really good and a nice change to the lecture format.

Assignments. Når man får lov til selv at sidde og anvende det der er blevet behandlet i forlæsningerne.

Comment

I really liked that the exam assignments used data from published papers so that the data is actually usable (more than made up data) and is a great way to reflect the result of the analysis to actual experiments

Jeg får meget ud af at side og kode selv.

forelæsningerne

You learn a lot from the assignments, sometimes i would be quite confused after a lecture, but then working with it in the assignments helped a lot.

Portfolio work, in class explanations, especially the illustrations (graphs, videos)

Doing the portfolio assignments / exercises

Despite my criticisms (slightly unsafe learning environment, tedious and vague assignments, and not fulfilling the potential of the Cryptpad), Lau is an amazing lecturer, and the course has been a pleasure (I even liked the jokes)!

PCA

At arbejde med ægte data i vores assignments, fra eksperimenter som er relevante for studiet har været rigtig fedt! Dog har det til tider føltes som at blive kastet ud af et fly uden faldskærm, når vi har skulle gøre matematik til kode, og lære et helt nyt programmerings-sprog ovenikøbet. Til en anden gang vil det være en stor hjælp, hvis vi som studerende havde mulighed for at skrive noter undervejs i forelæsningen, da det er nemt at blive hængt af. Men dette er svært, når man ikke har adgang til undervisningslides ved timens start. Det er forståeligt, at det som underviser kan være frustrerende at tale til en masse elever, som har øjnene fikseret på skærmen – men når det går stærkt, er det rigtig udfordrende ikke at kunne markere, hvornår man er blevet hængt af, og hvad man skal kigge på igen, når slides er blevet offentliggjort. Jeg vil igen pointere, at jeg har en fornemmelse af, at os studerende har lært rigtig meget dette semester – men det er gået meget stærkt af og til, hvor jeg kunne se, at udbyttet fra kuret vil have været endnu bedre, med denne lille ændring.

Portfolio assignments and Lydia

The assignments was very good at teaching us the basics of machine learning algorithms

Assignments helt klart, fordi man har kunnet få stoffet under huden og været nødt til at forstå materialet. Forelæsningerne kan hurtigt blive for abstrakte. Idéelt set ville jeg gerne ønske om flere assignments, men i de uger hvor vi havde assignments, havde jeg tilgængæld også alt alt for travlt, da der oftest også var ca. 250 siders læsning ved siden af, hvilket langt fra var muligt.

individual study.

I especially enjoyed getting hands on learning by doing relevant portfolios that build on what we had learned during the lectures. The classes with Lydia and doing exercises are also a great way for me to learn and something I enjoyed.

Assignments and work with the study group

My takeaways

- I need to make live coding more approachable
- I need to make the first two assignments have more clear wording (but overall assignment structure is liked)
- I will keep asking “random” people questions, but I can work even more on the communication
- Maybe have a linear algebra class (for the math)
- Not getting slides was mentioned less than expected
- UCloud?